



INSTITUT INTERNATIONAL DU FROID
INTERNATIONAL INSTITUTE OF REFRIGERATION

Sustainable Cooling Technologies for Data Centers

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2025.7.7



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- Data center supports artificial intelligence, IoT, big data analysis.....

Artificial intelligence

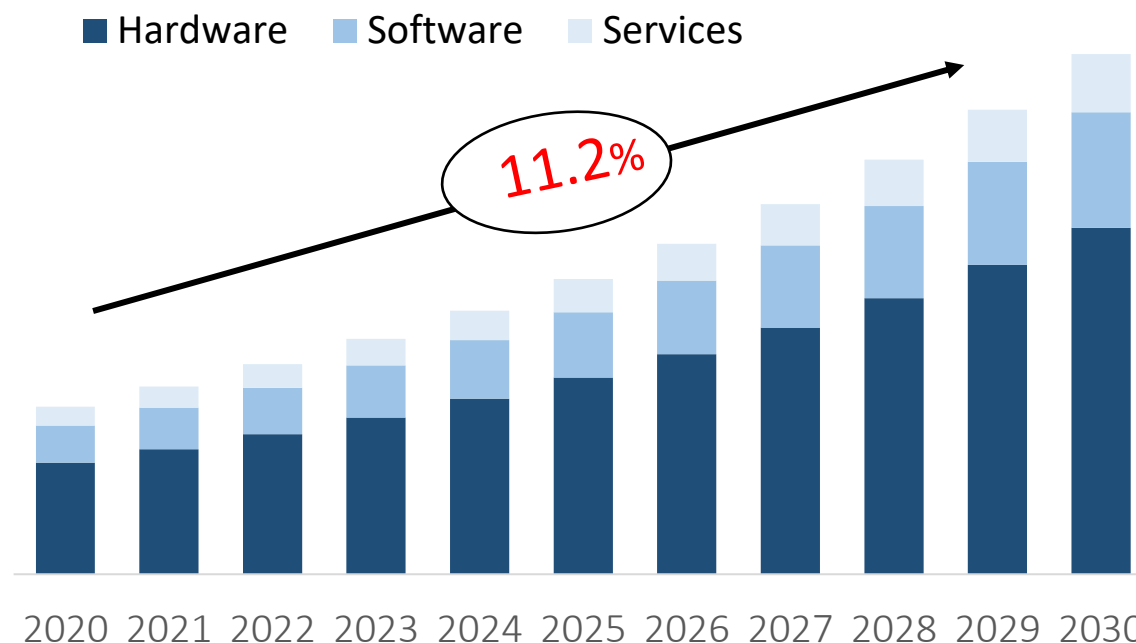


Industrial IoT



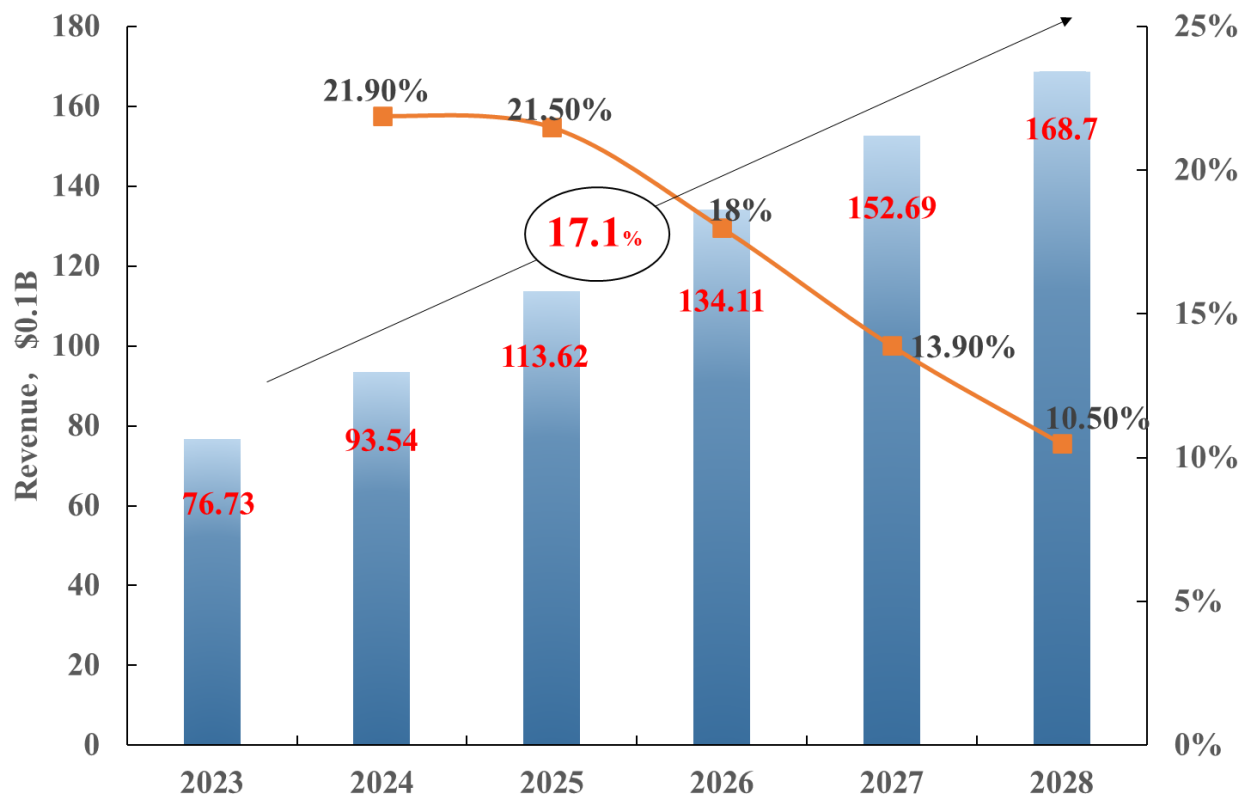
Big data analysis

Data center market composition

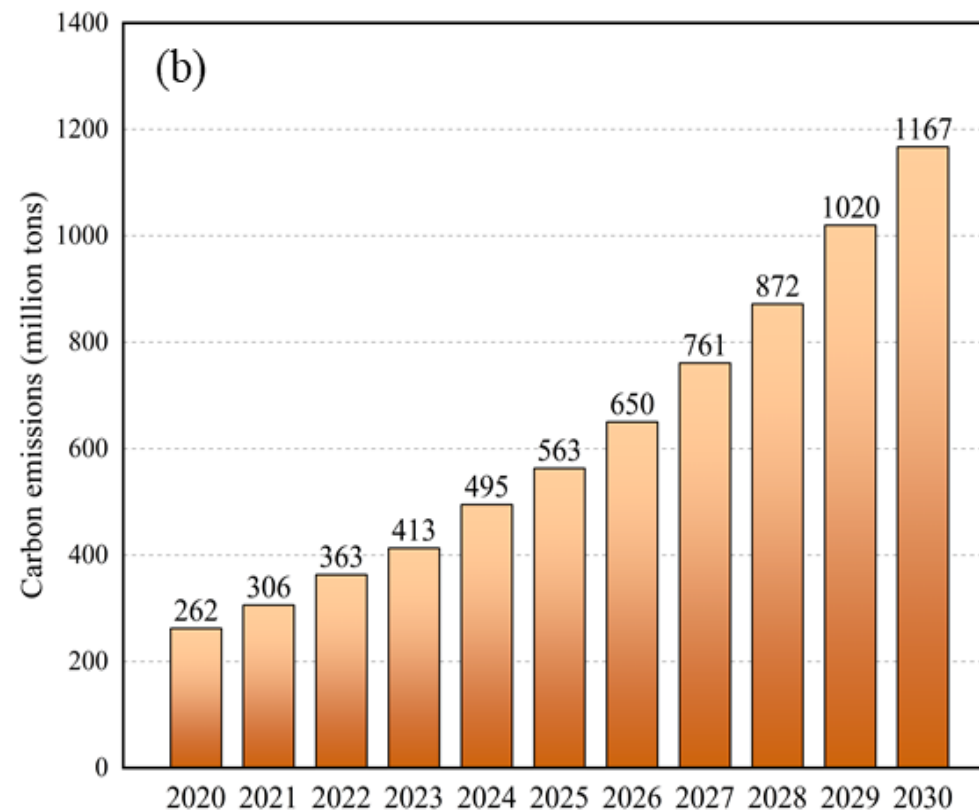


- Market projections indicate a compound annual growth rate of 11.2%

Global data center cooling market size



Global carbon emissions from data centers



**Increased utilization of data centers →
Elevated energy consumption and carbon emissions**

1

| Background

2

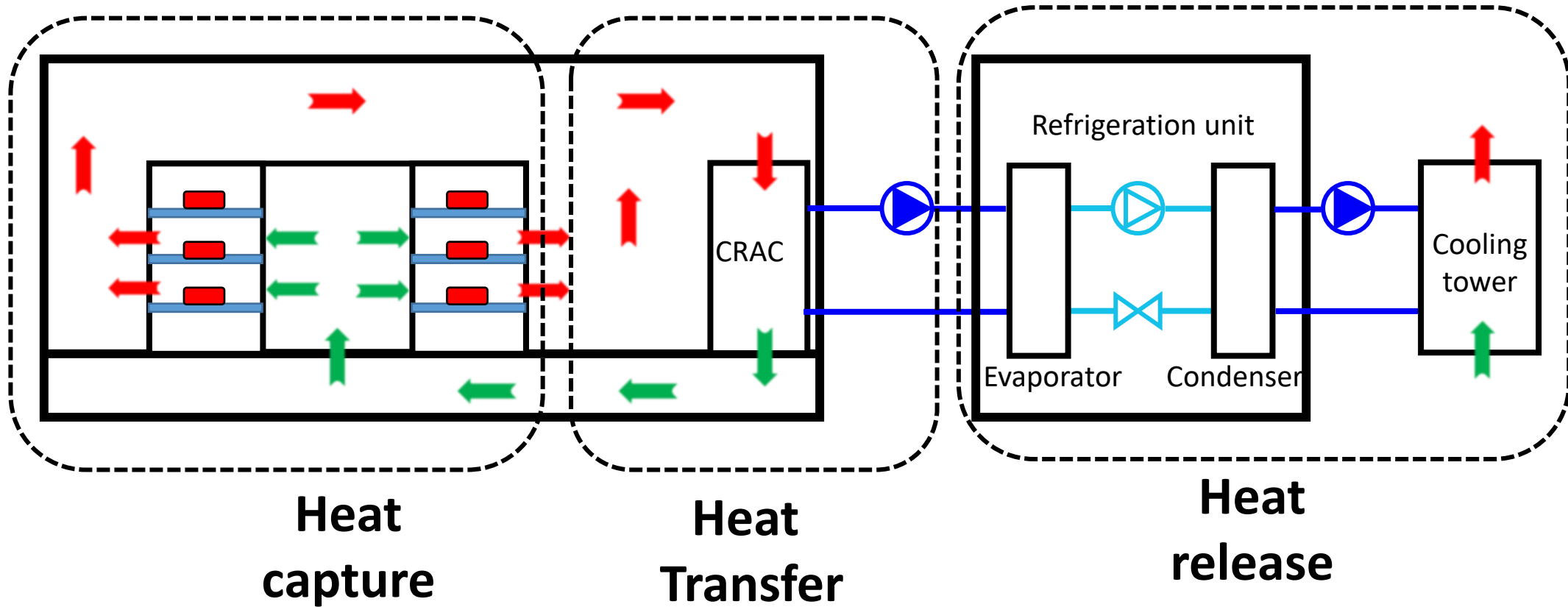
| **Principle**

3

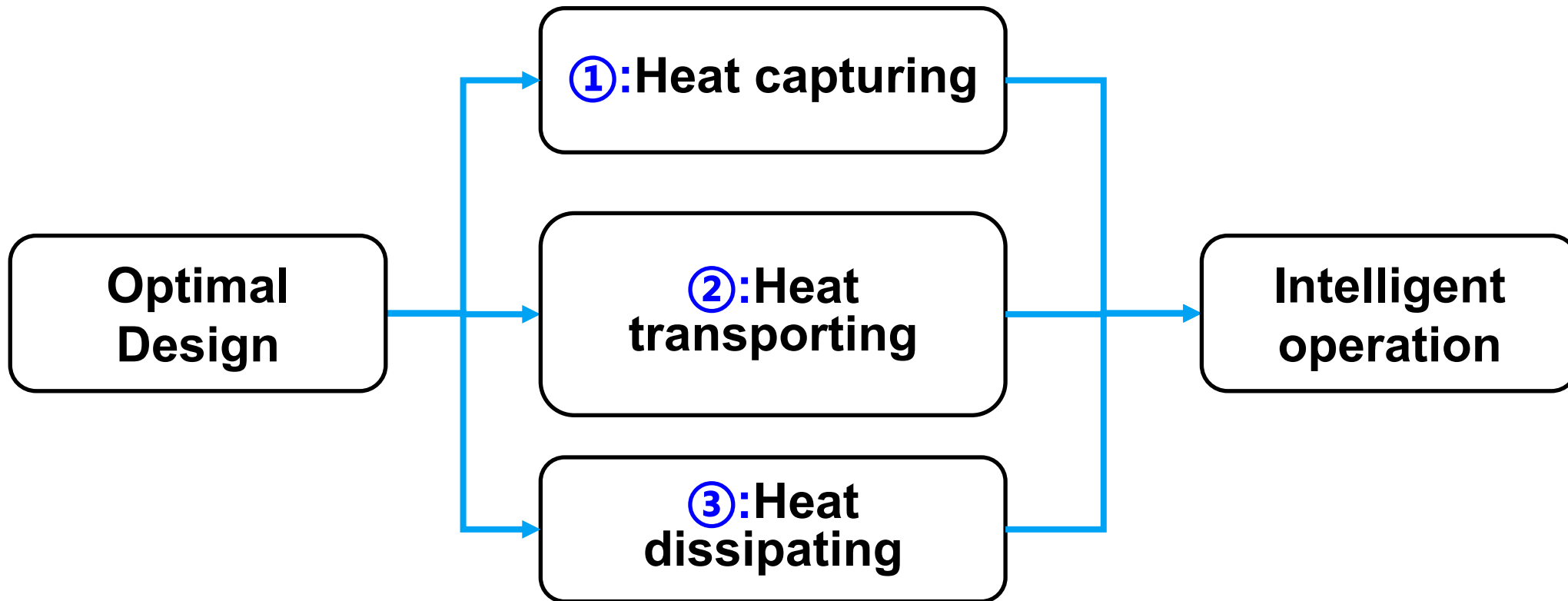
| Sustainable Cooling Technology

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- Pathways for sustainable cooling of data centers



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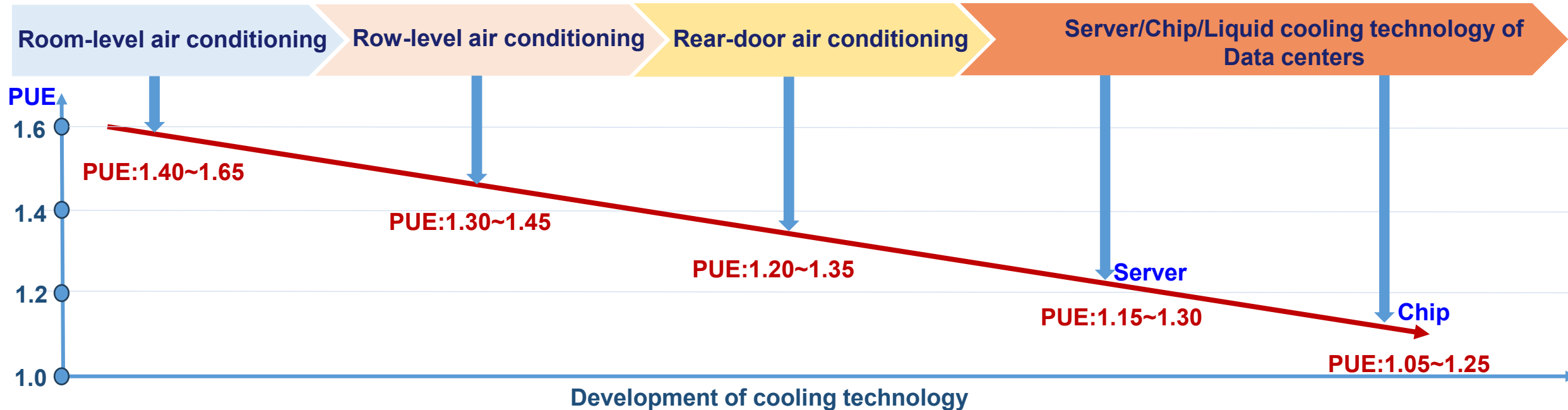
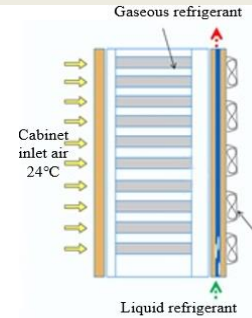
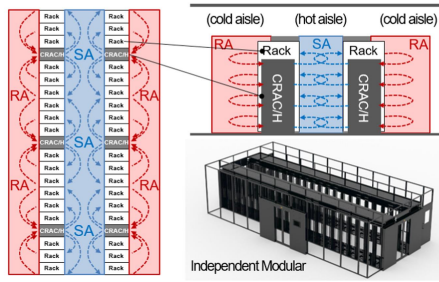
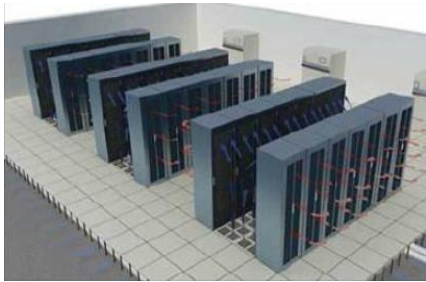
| **Sustainable Cooling Technology**

4

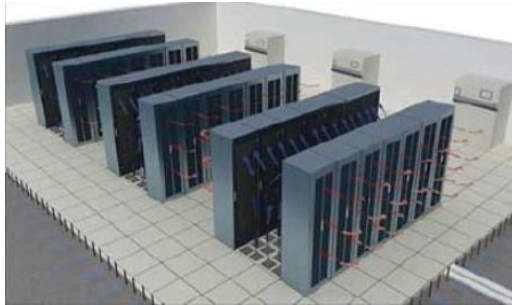
| Conclusions

3.1 Heat capturing

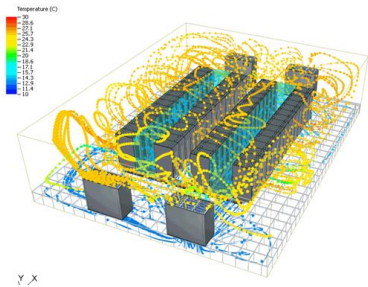
Target: 1) Decrease the heat exergy degradation (by heat transfer or mixture), and
2) Enhance the heat transfer intensity



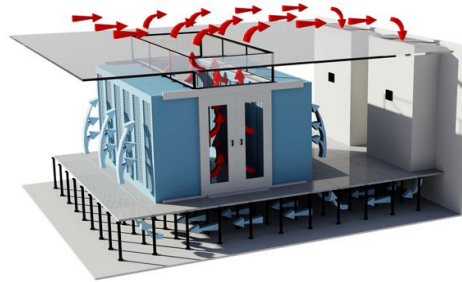
① Room-level air conditioning



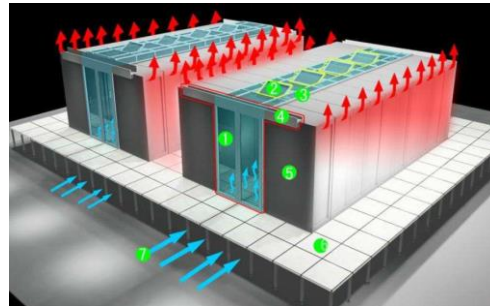
Hot aisle/cold aisle
without containment



Airflow management

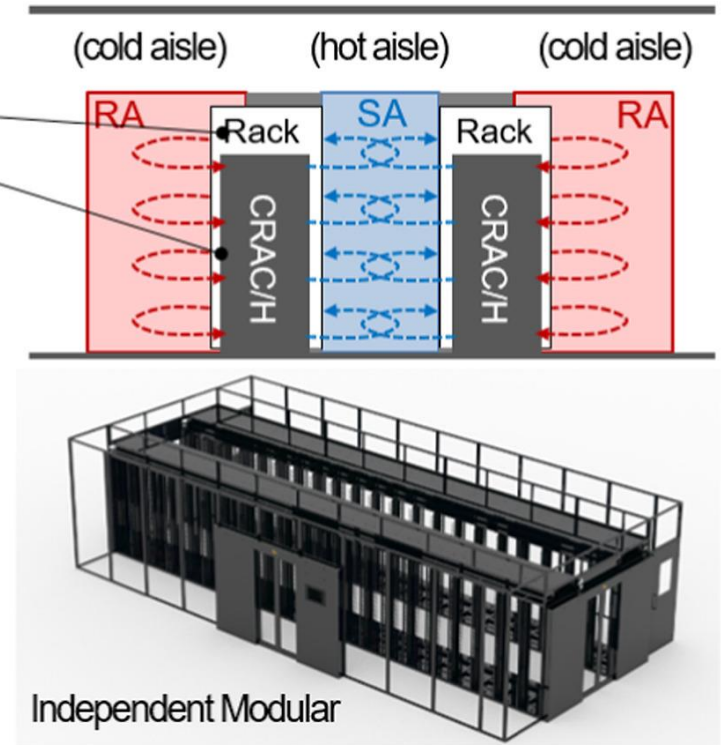
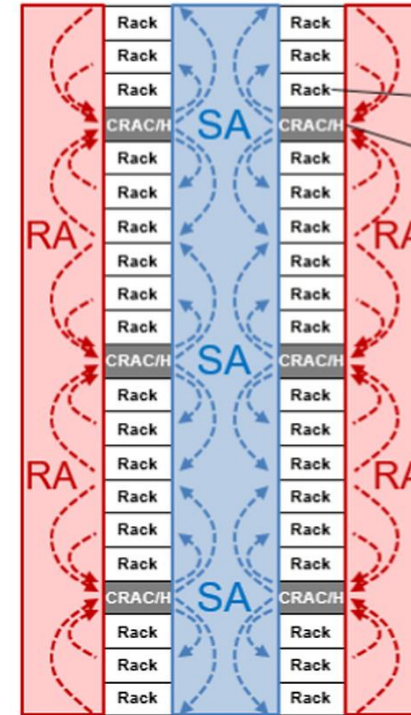


Underfloor air supply
with hot aisle closure



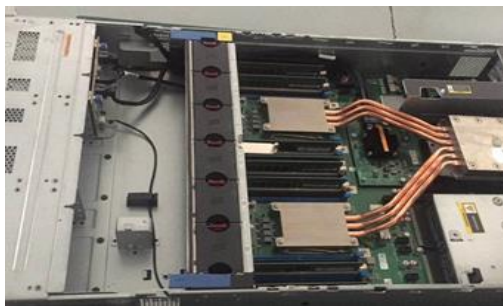
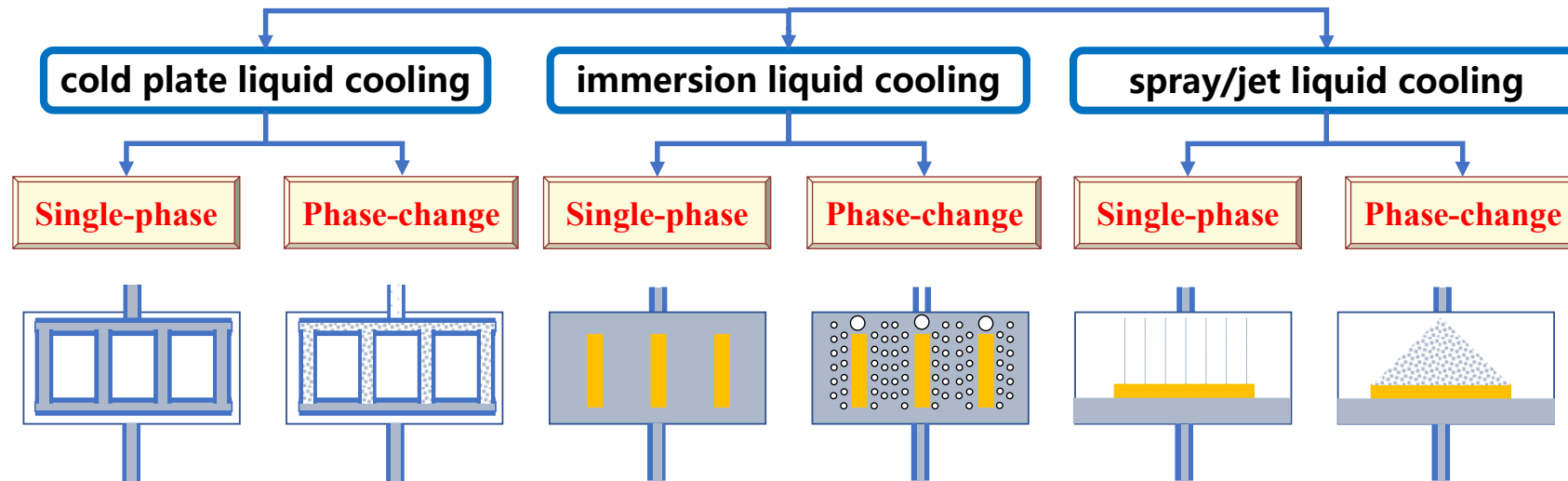
Underfloor air supply
with cold aisle closure

② Row-level air conditioning



minimizes airflow distance, enables precise demand-driven cooling,
and consequently enhances energy efficiency

③ Liquid cooling technology of data centers



Cold plate

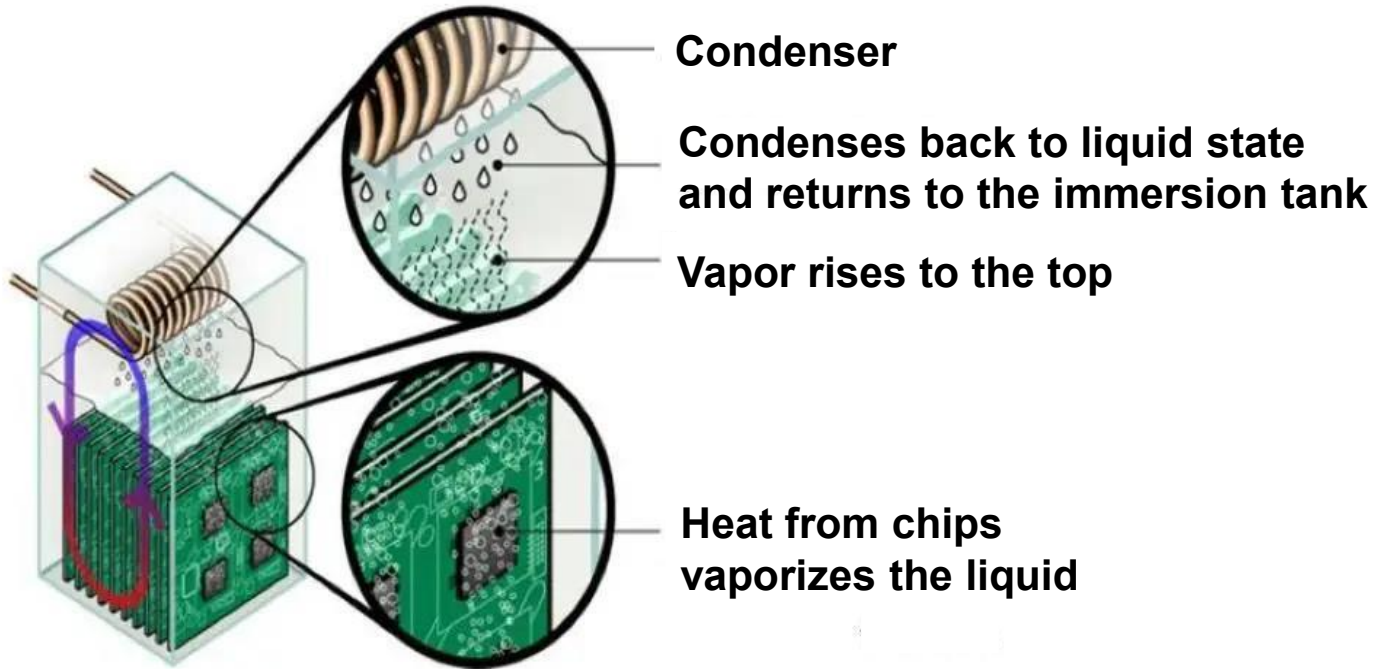


Immersion liquid cooling



Spray liquid cooling

Two-phase immersion liquid cooling

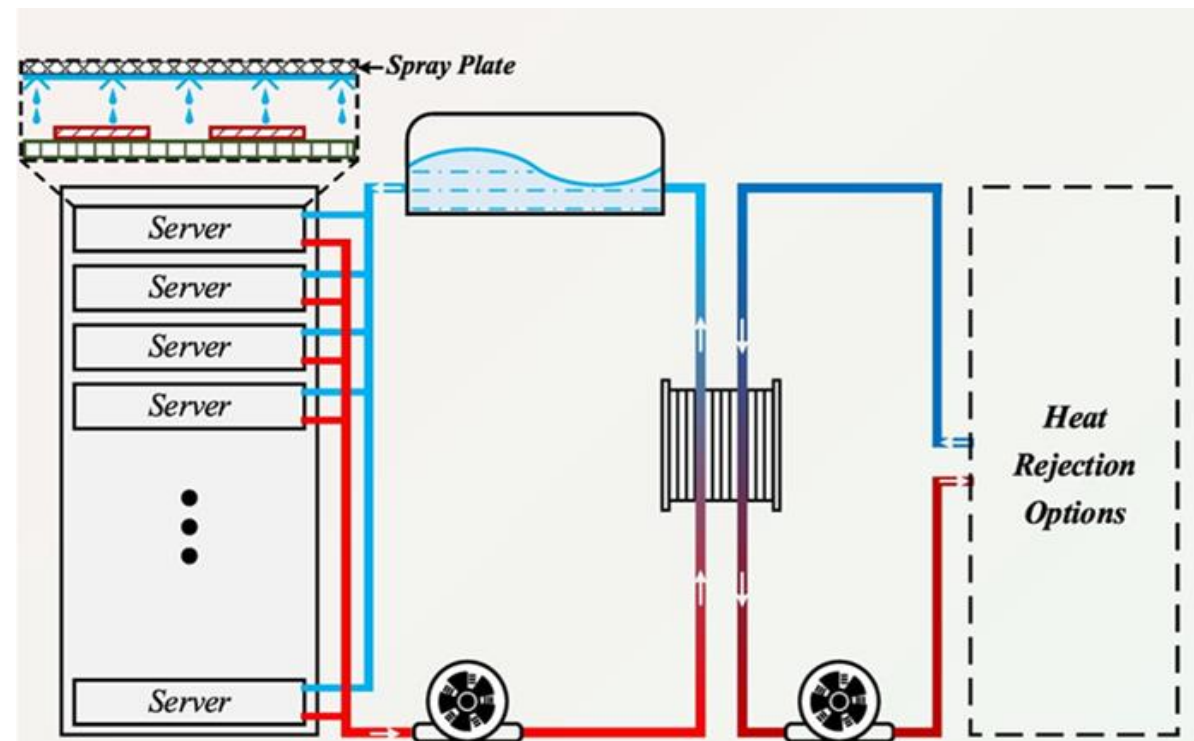


- Eliminating airflow-dependent cooling infrastructure
- Achieving near-theoretical energy efficiency limits
- Supporting ultra-high power density deployments

Spray/Jet liquid cooling

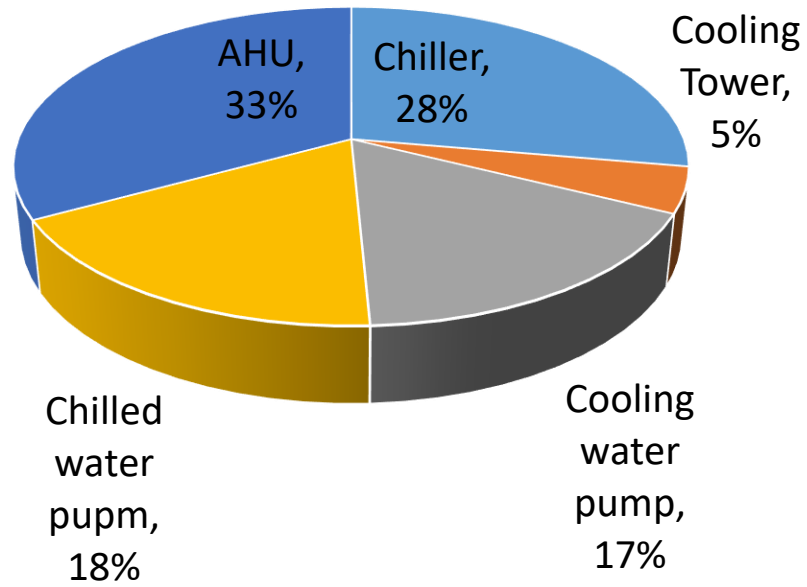


- Targeted spraying based on localized heat flux density
- **Reduces coolant consumption**
- Decreases structural floor loading



Comparison of different Liquid cooling technologies

	Cold plate	Immersion	Spray/Jet
PUE	★★★★☆	★★★★★★	★★★★☆
Capital Expenditure	★★★★☆	★★☆☆☆	★★★☆☆
Space Utilization Efficiency	★★★★☆	★★★★★★	★★★★☆
Technology Maturity	★★★★★	★★★☆☆	★★☆☆☆
Max Thermal Load per Rack	★★★☆☆	★★★★★★	★★★★☆
Retrofit Compatibility	★★★★☆	★☆☆☆☆	★★☆☆☆



The energy consumption proportion of Heat transport within data centers' annual total energy usage is continuously rising

Target: 1) **Minimize transport power required**, and
2) **Use environmental friendly coolant**

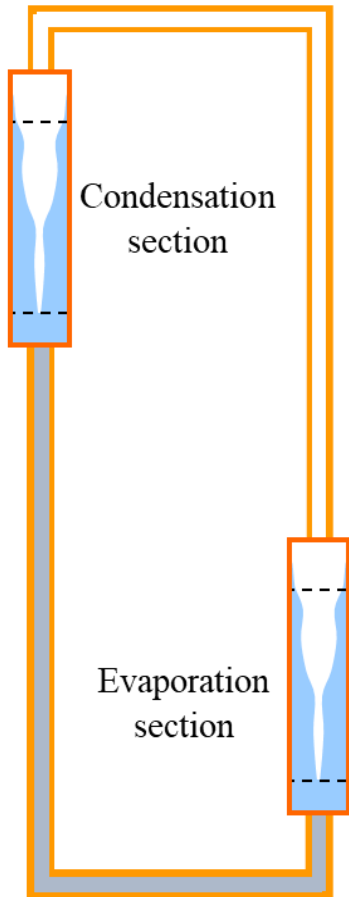
$$Q = \dot{m}C_p(T_i - T_o)$$

→ **Larger heat transfer temperature differentials** increase exergy destruction
→ **Lower flow rates** reduce fluid transport energy consumption

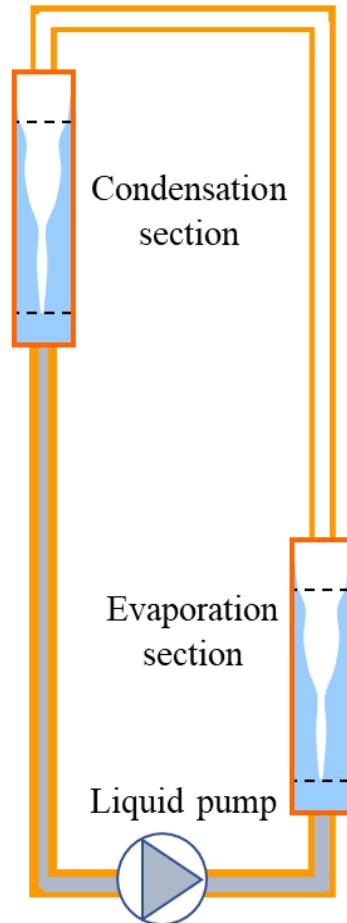
$$P = \dot{m}\Delta H$$

→ **Refrigerants' lower viscosity** versus water, coupled with vapor-liquid density differences, enable reduced transport energy
→ **Higher fluid density** permits smaller piping diameters, conserving space

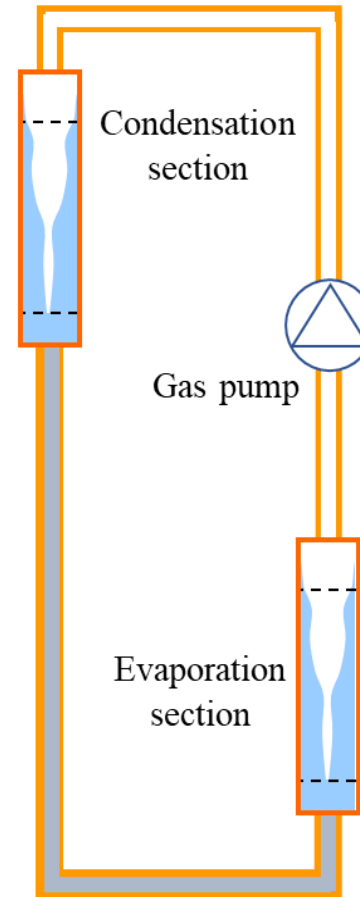
Heat pipe loop



(a) Gravity-driven(thermosyphon)



(b) Liquid pump driven



(c) Gas pump driven

Gravity-driven(thermosyphon)

Utilizes gravity potential and density differential as circulation driving force.

Liquid pump driven

Utilizes liquid pump as circulation driving force; cavitation prevention required.

Gas pump driven

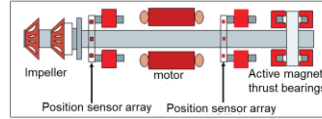
Utilizes vapor compressor as circulation driving force; liquid slugging prevention required.

Integrated system architecture



⑤ High-Efficiency Chillers

- **Magnetic bearing variable frequency** drive chillers
- Water/air-cooled VFD chillers
- Low condenser water temperature operation



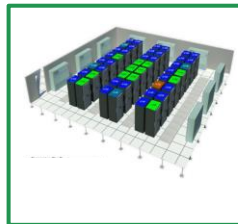
⑥ High-Efficiency Customized Terminal Units

- Electronically commutated (EC) fans
- Fresh air / full recirculation mode



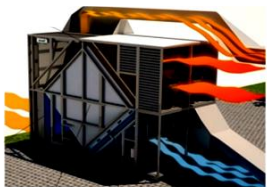
⑦ Packaged Cooling Plant System

- Prefabricated modular units
- Unitary integrated cooling plant
- Integrated building management system



⑧ AI-Driven Energy Management Platform

- All-variable speed drive configuration
- Free cooling; redundant hot standby

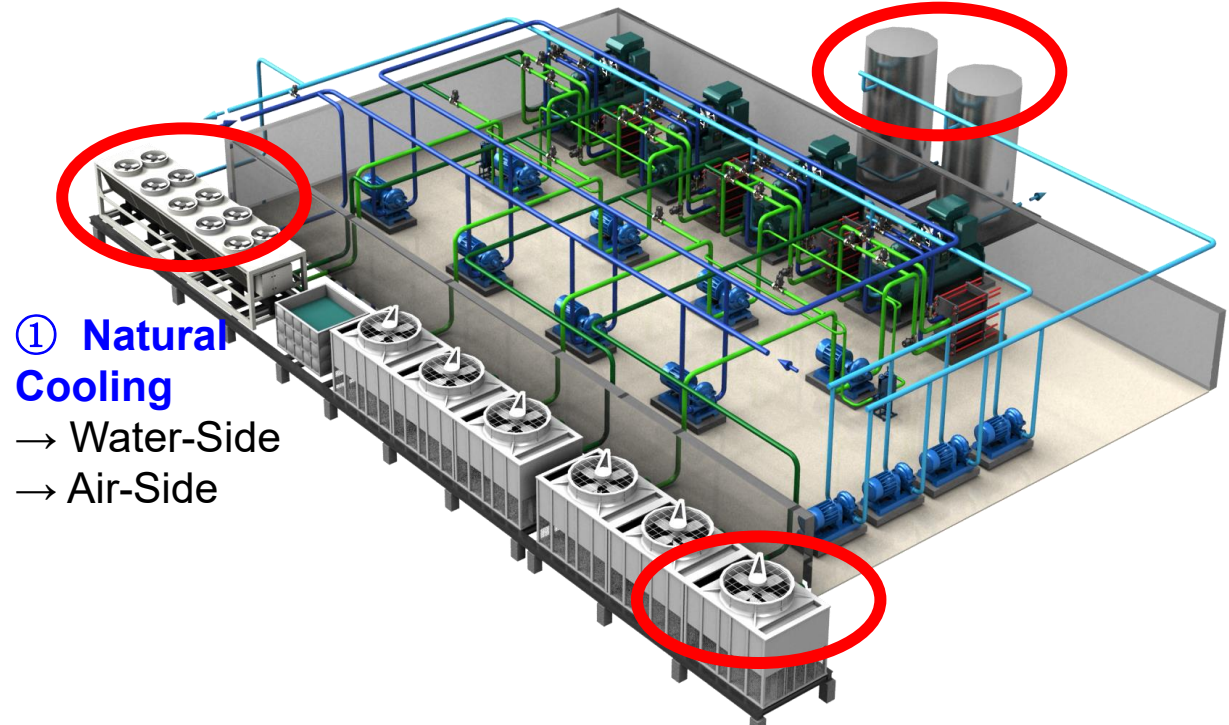


③ Evaporative cooling

- reduction of cooling source temperature

② Thermal Energy Storage

- Ice-based thermal energy storage
- Water-based sensible thermal energy storage



① Natural Cooling

- Water-Side
- Air-Side

④ Waste Heat Recovery

- Condenser Heat Recovery System
- Total Heat Recovery

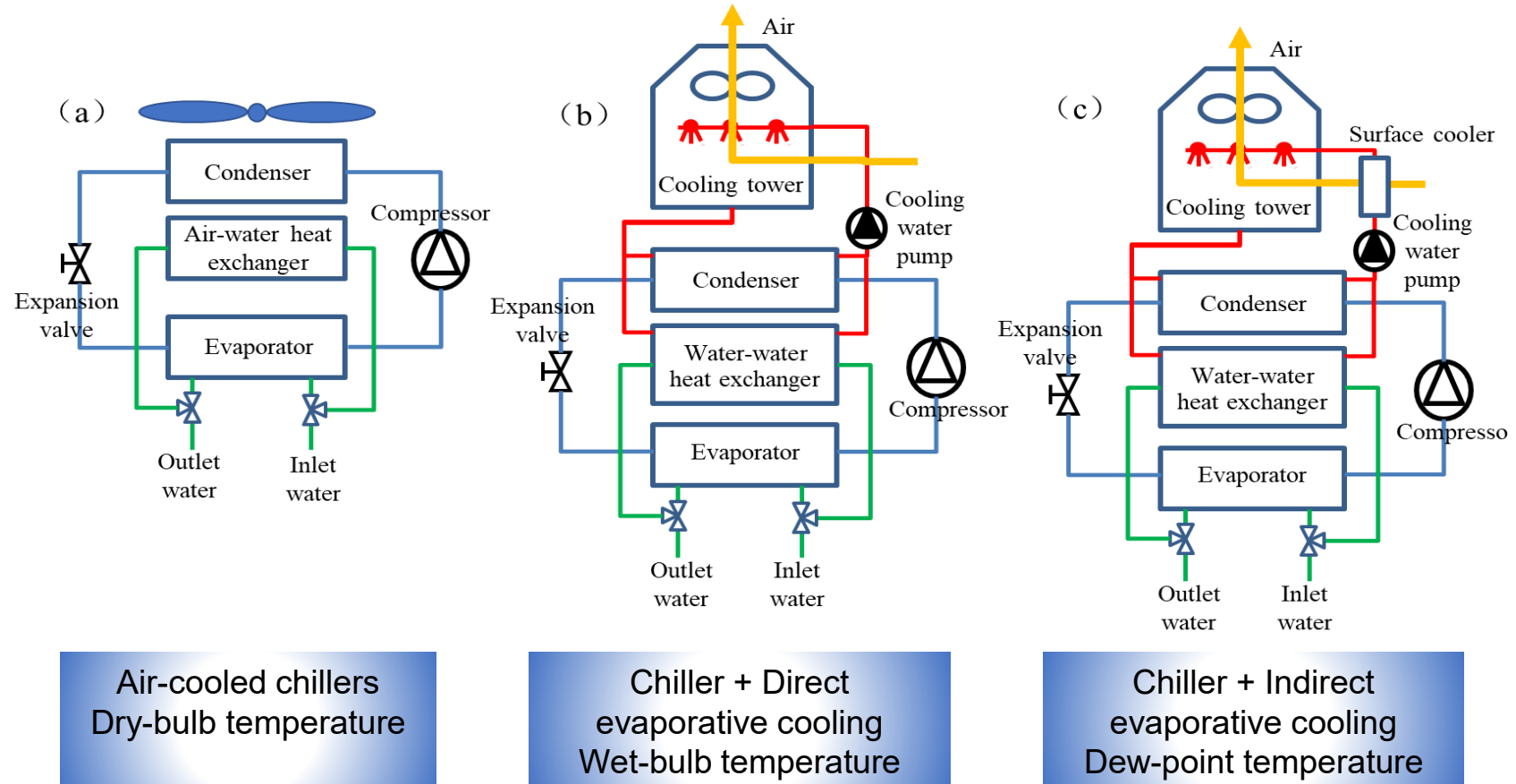
① Natural cooling

Types of natural cooling system

Classification	Type
Air-side	Direct Indirect
Water-side	Direct water-cooling Air-cooling Cooling tower /evaporative cooling
Heat pipe	By driving force By system By outdoor cooling

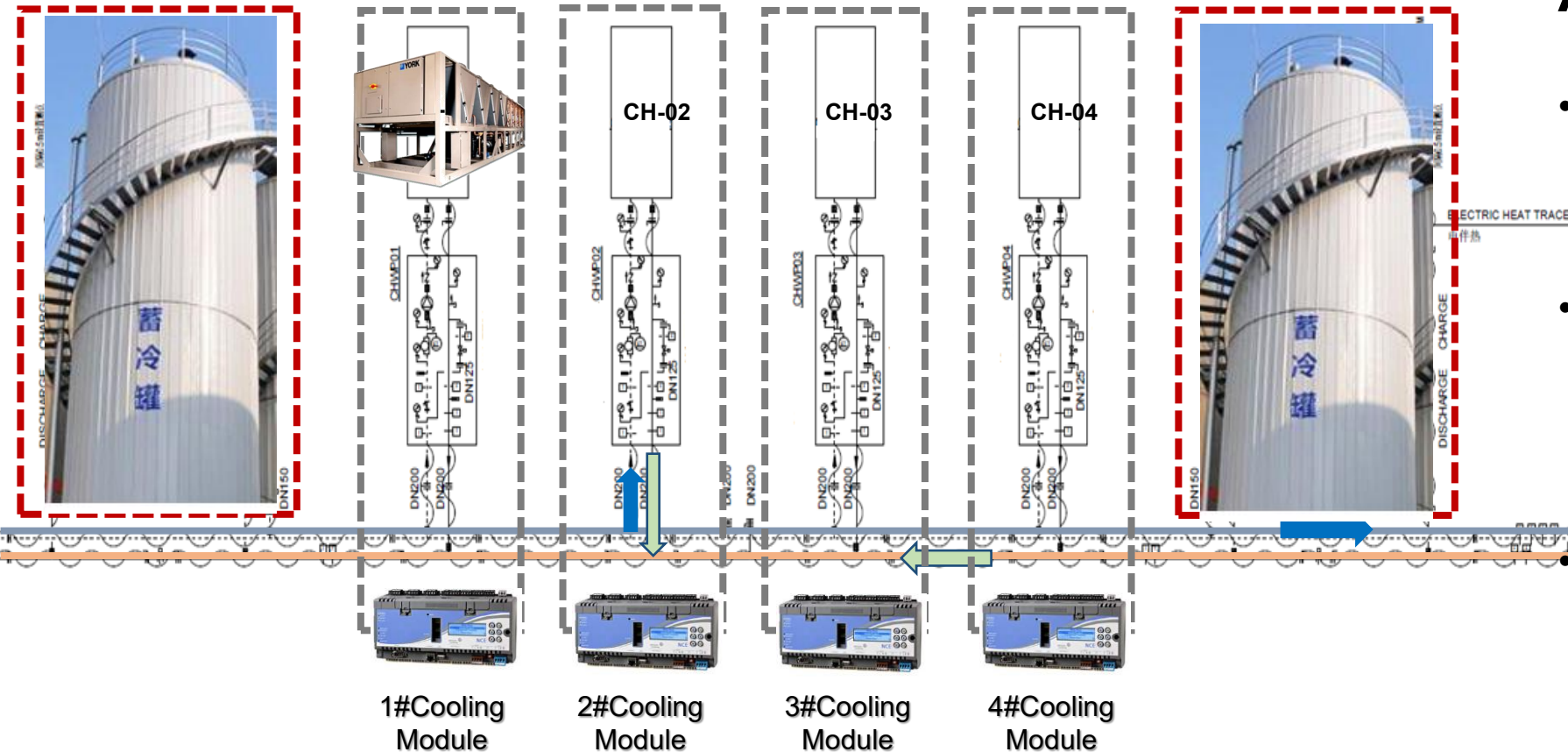
Advantages:

- Environmental Sustainability
- Economic Viability



Active/natural cooling integrated systems

② Data Center Cooling Thermal Storage

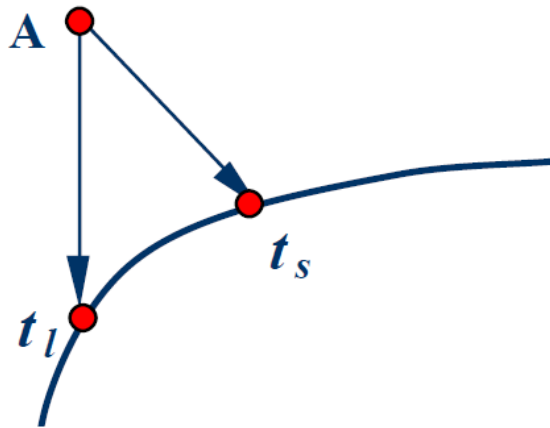


Advantages:

- **Increasing Cooling Capacity:** Enhances the operational resilience of data center cooling systems
- **Customer-Side Energy Storage:** Improves grid reliability and increases renewable energy utilization rate
- **Time-of-Use Electricity Pricing:** Reduces operational costs

③ Evaporative cooling: reduction of cooling source temperature

Typical Evaporative Cooling Heat Transfer Process



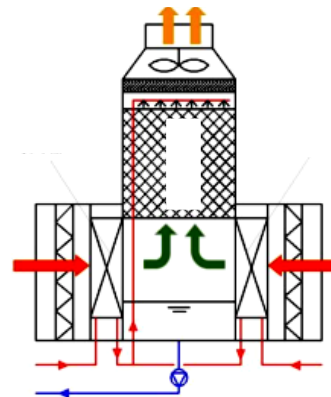
- Heat removal via water evaporation
- Direct evaporative cooling (t_s) / Dew-point evaporative cooling (t_l)
- Generates chilled air/water or condenses refrigerant (evaporative condensation)



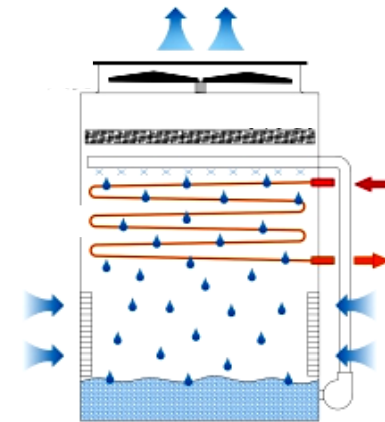
Internally-cooled indirect evaporative air handling unit



Externally-cooled indirect evaporative air handling unit



Indirect evaporative chiller



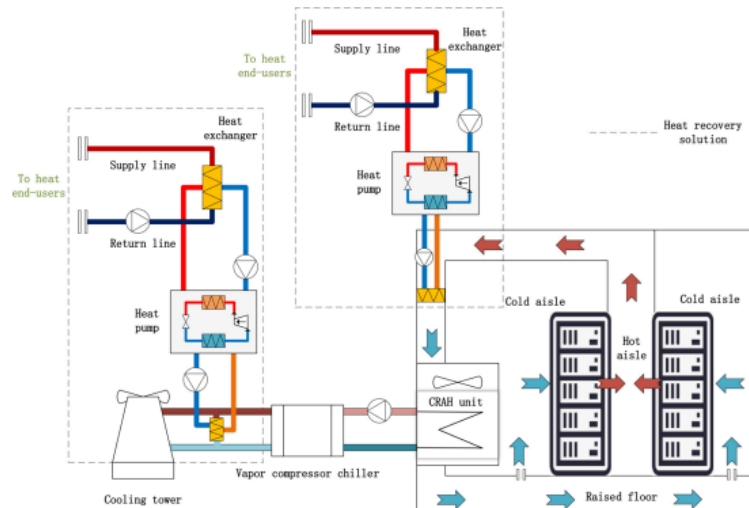
Evaporative condenser unit

④ Data Center Waste Heat Recovery Technologies

Various heat sources

Air Cooling Thermal Sources: High-temperature exhaust air, condenser water, chilled water

Liquid Cooling Thermal Sources: Coolant return flow, process cooling liquid



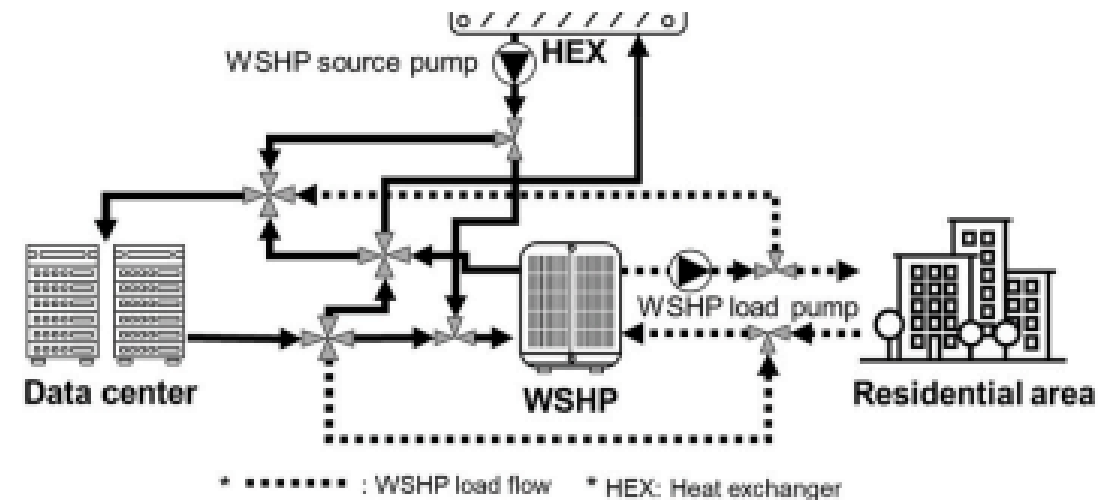
Air Cooling Thermal Sources

Various heat sinks

District Heating: Supplies hot water/space heating

Organic Rankine Cycle (ORC) Power Generation: Directly offsets electricity consumption

Supplementary Cooling: Waste heat-driven refrigeration compensating cooling load



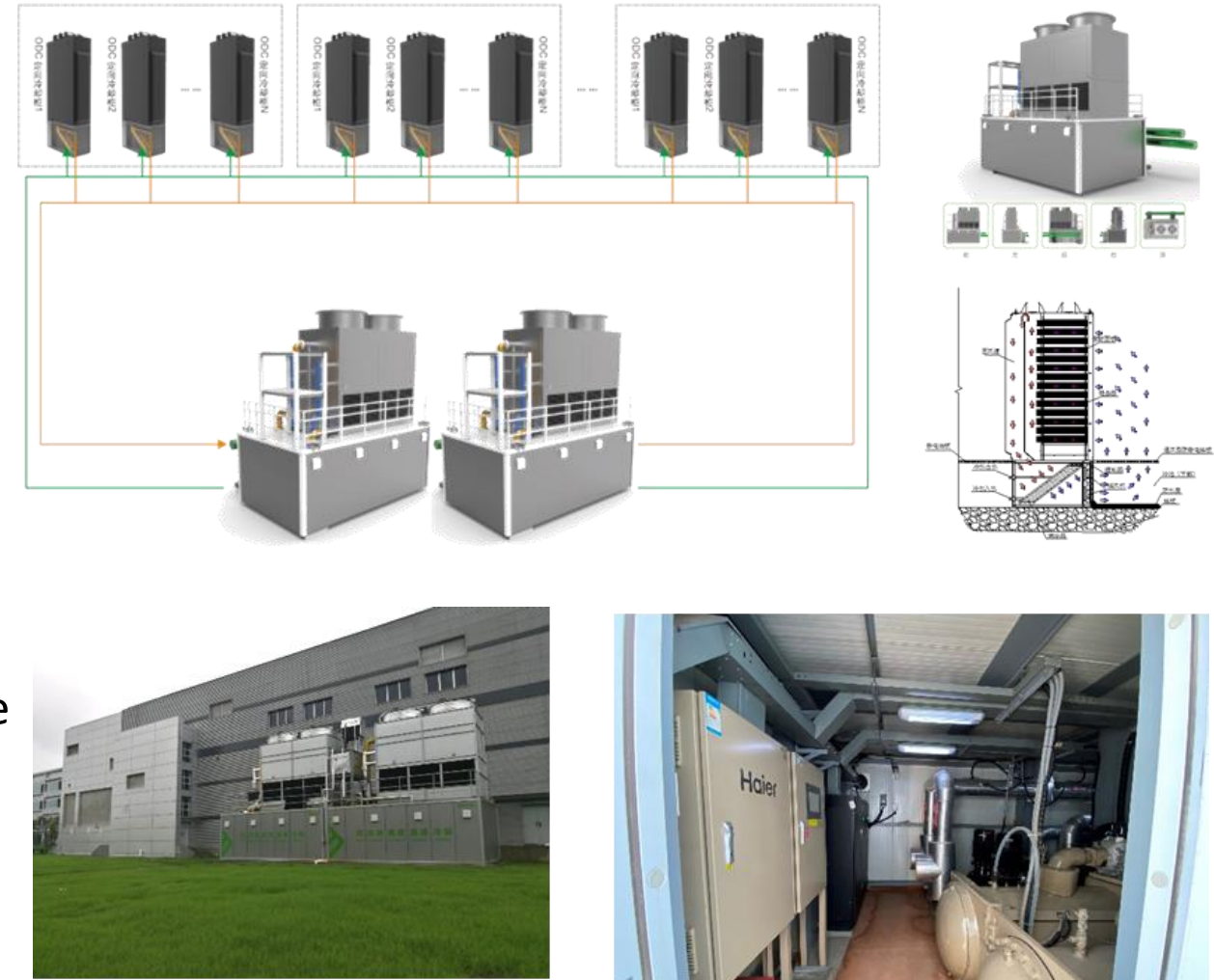
District Heating

Integrated Cooling Plant- Qingpu, Shanghai

Cooling System Architecture:

- **Dual-loop system design** with two cooling plants operating in parallel to provide mutual hot standby capability
- Implementation of **closed-circuit cooling towers** enabling valve-free **free cooling** operation
- **Customized rack-level cooling terminals with 22 C elevated supply temperature and balanced hot aisle closure**

Global Coefficient of Performance (GCOP) for the entire cooling system and **Annual Coefficient of Performance (ACOP)** for the integrated cooling plants are **6.95** and **8.77**

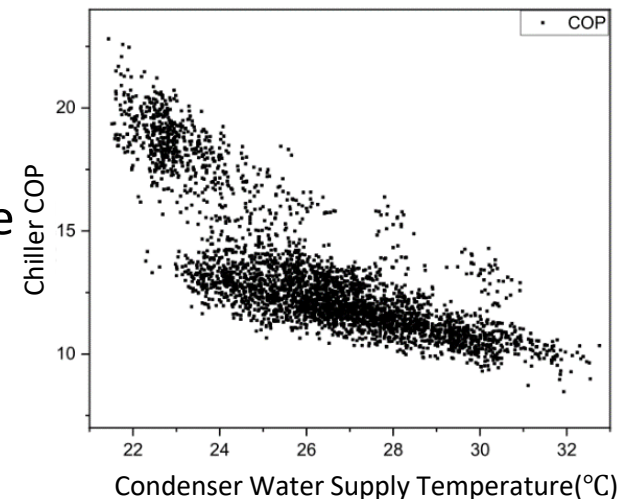
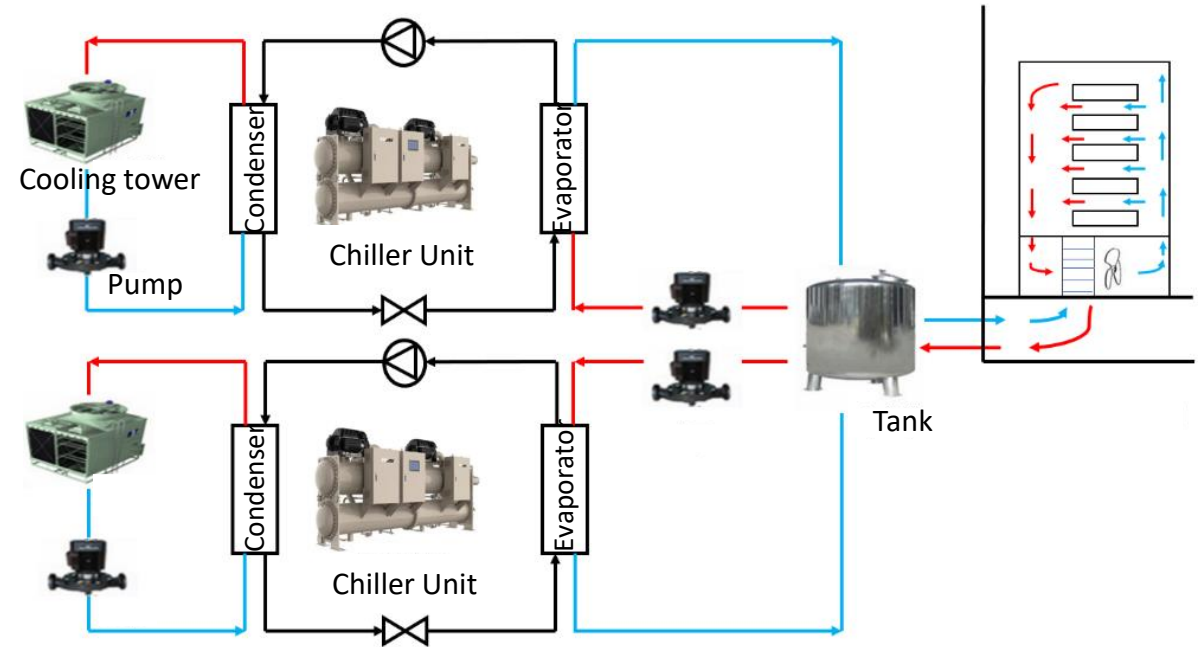


Magnetic Bearing Centrifugal Chillers - Beijing

Cooling System Architecture:

- The cooling system employs **two magnetic bearing centrifugal chillers**
- **Rack-level cooling** terminals minimizing indoor energy consumption
- **A custom-engineered oil-based heat transfer fluid** with enhanced thermal efficiency and IT equipment safety

Magnetic bearing chillers achieved significant **noise reduction (≤ 75 dB(A))**. The calculated seasonal average **Coefficient of Performance (COP)** reached **12.69**



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- The rapid expansion of data centers drives massive electricity consumption, resulting in significant greenhouse gas (GHG) emissions
- To address the intensified heat generation from super servers, more powerful cooling systems are imperative
- Developing sustainable cooling solutions demands a comprehensive approach — from efficient heat capturing to low-energy heat transport and optimized heat dissipation mechanisms

Thanks